

STUDYING THE EFFECTS OF CATIONICSURFACTANT MODIFICATION ON THE ACID BLUE 25 FILTRATION CAPABILITIES OF ZEOLITE

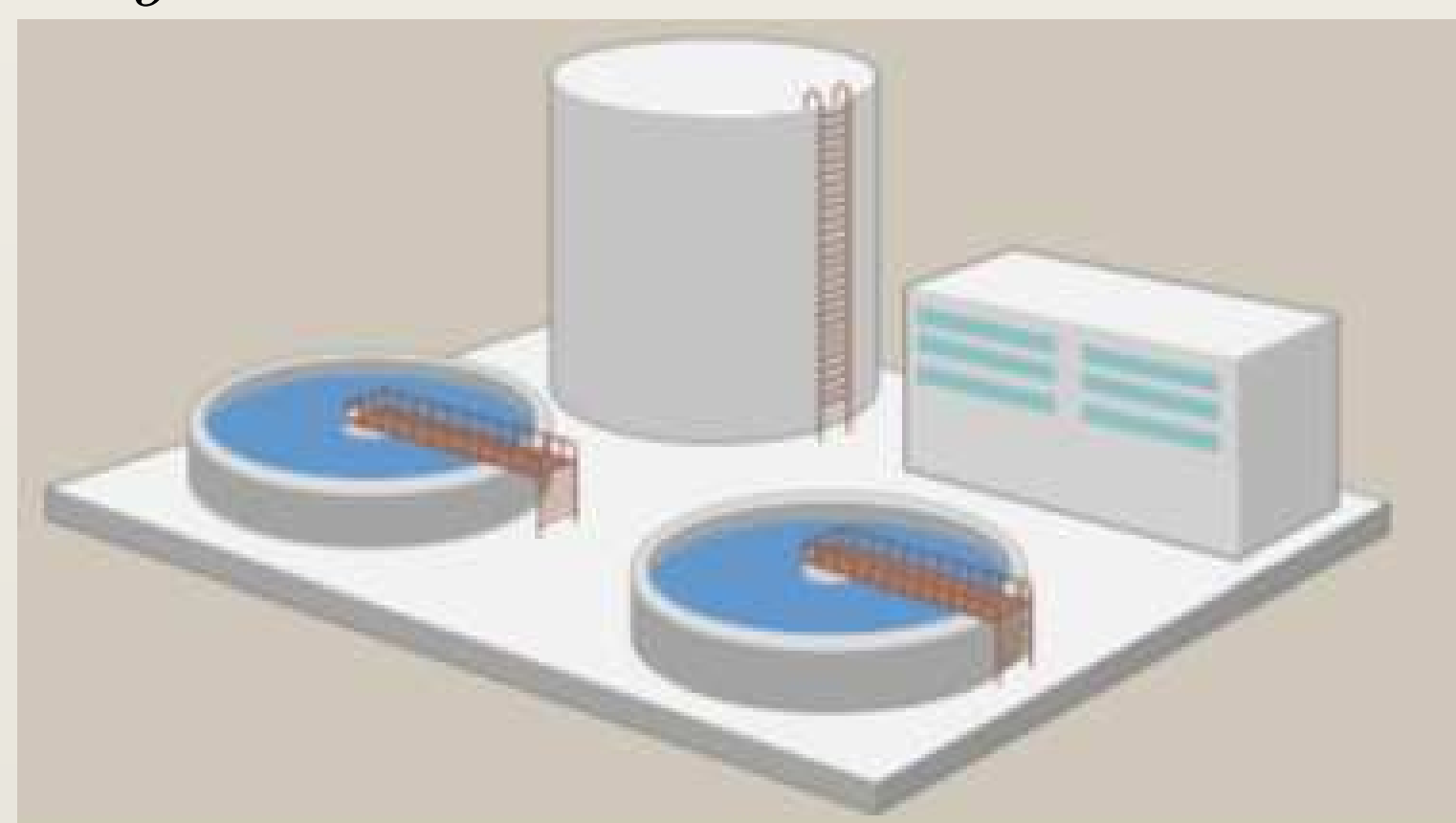
MATERIALS SCIENCE

Intro/Background

Acid blue 25 is a common dye and anionic pollutant that is most often found in wastewater from plants in the textile industry (Ghodbane & Hamddaoui, 2009). Upon ingestion, it can cause gastrointestinal pains such as nausea and vomiting. Due to its prevalence and the harm posed by the chemical, acid blue 25—like many other anionic pollutants that contaminate our drinking water—is currently being researched for removal.

One popular modern method for removing pollutants in wastewater is using the porous material zeolite.

Diagram - Wastewater Treatment Plant

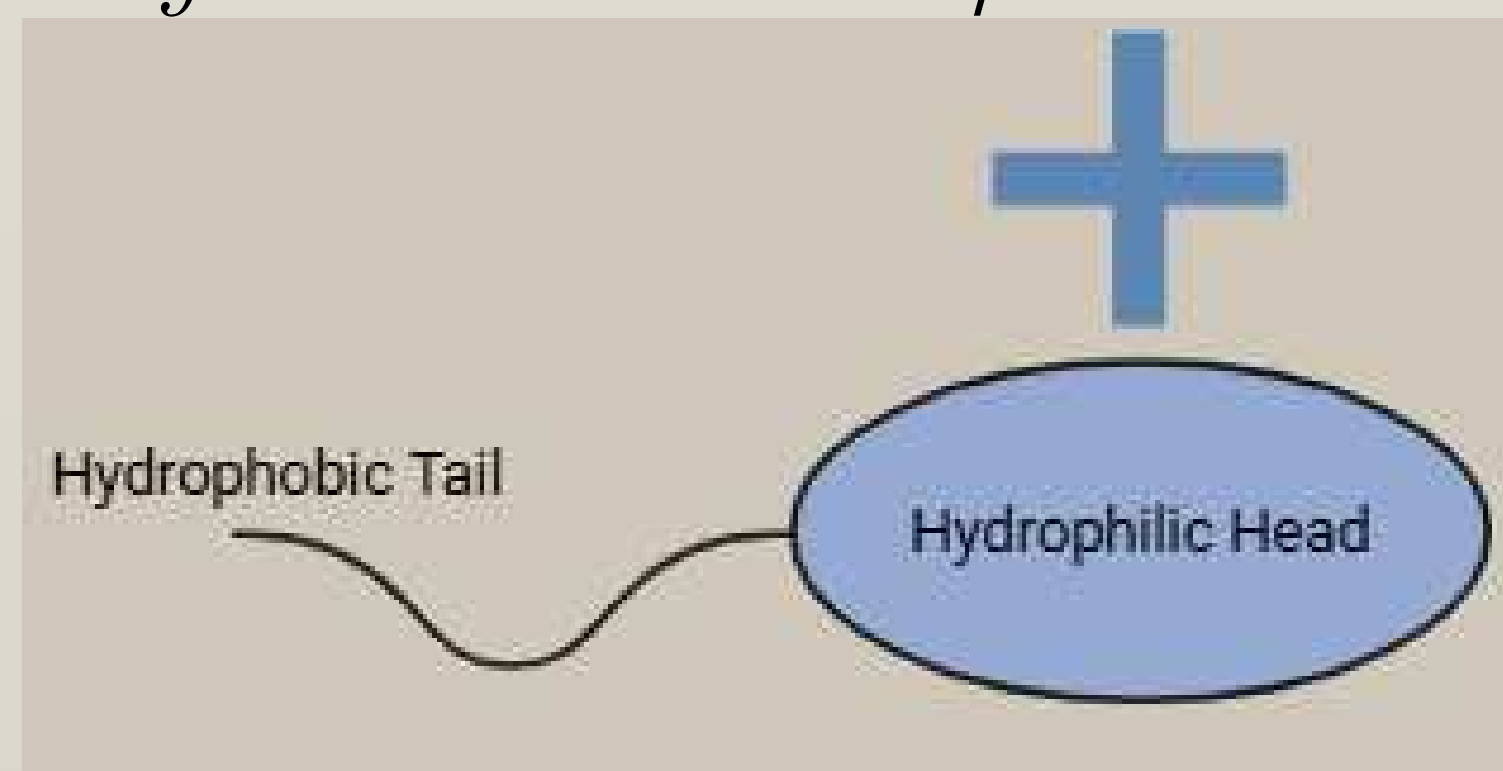


Problem + Hypothesis

Filtration of soluble anionic surfactants is still imperfect as physical separation does not work. Current zeolite filtration has a primary focus on removing solid pollutants, like microplastics, so it is not quite as effective against dyes.

The formulated hypothesis was that modifying zeolite with the cationic surfactant (CTAB) would increase the acid blue 25 filtration capabilities of the zeolite because, as past research supports, the positive charge of cationic surfactants would attract negatively charged anionic pollutants (Shen et al., 2021).

Diagram - Cationic Surfactant Structure



Methodology

Variables: IV was surfactant modification (modified, unmodified). DV was absorbance.

Control: Unmodified zeolite.

The main steps of my experiment were:

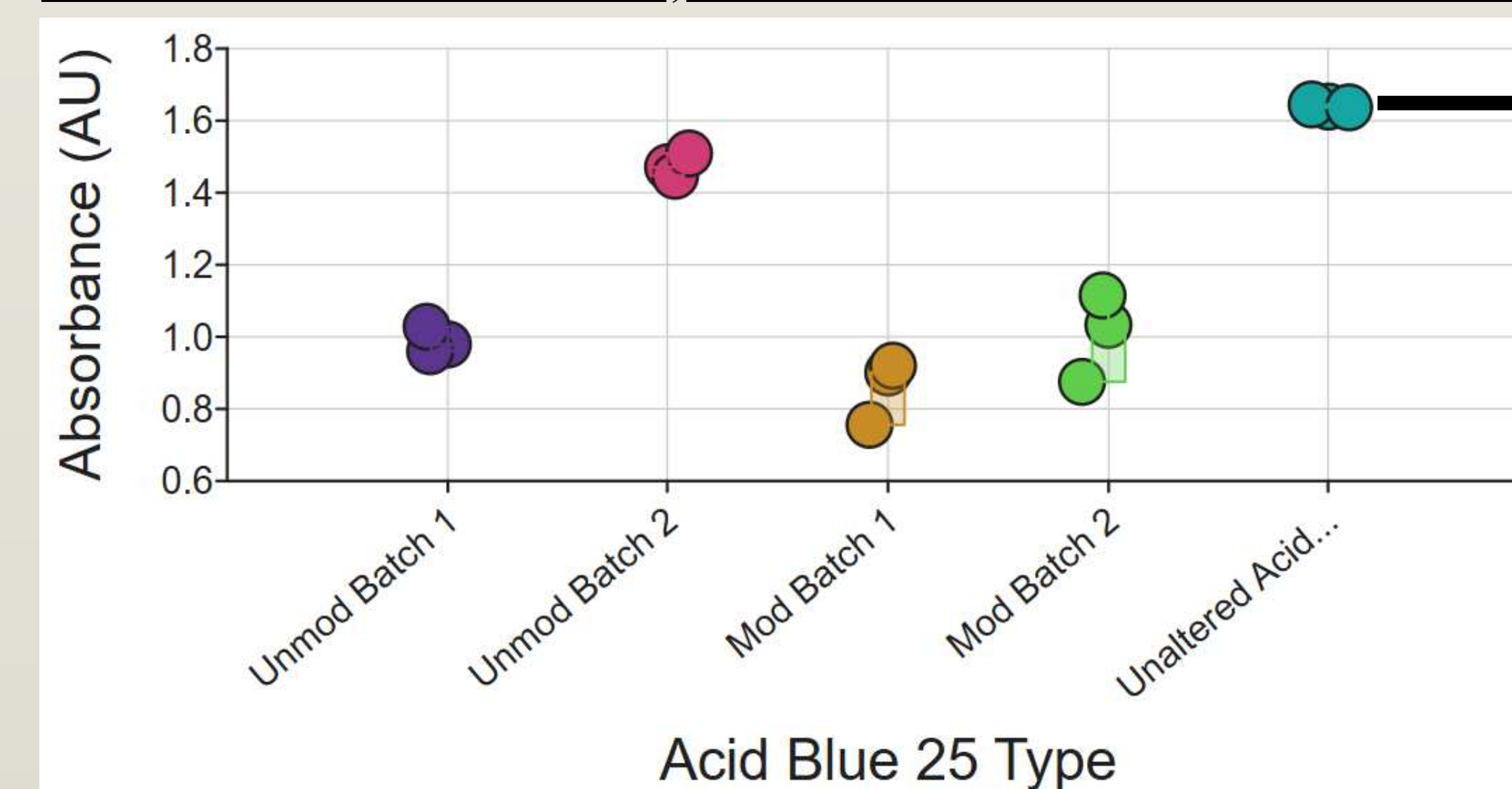
1. Zeolite synthesis - A mixture of non-carcinogenic chemicals were stirred together into a gel, hydrothermally treated, removed, dried, and calcined to create the zeolite (Shen et al., 2021).
2. Surfactant modification - Zeolite was modified via a 48 hour submersion in CTAB solution, which was inside a beaker being submerged in a constantly heated oil bath to ensure the CTAB did not revert to solid form. FTIR is then used to verify CTAB attachment. Skipped for control.
3. Batch adsorption tests - In two batches, a zeolite sample was submerged in an acid blue 25 solution under stirring, where the solution is extracted after each batch's time (~30 minutes) and stored for observations. Three trials were conducted per IV value.
4. Absorbance Measurements - The UV Spectrovis was used to determine the absorbance of each post-batch test solution for comparison against the unaltered acid blue 25 solution.

Diagram - Method Flowchart



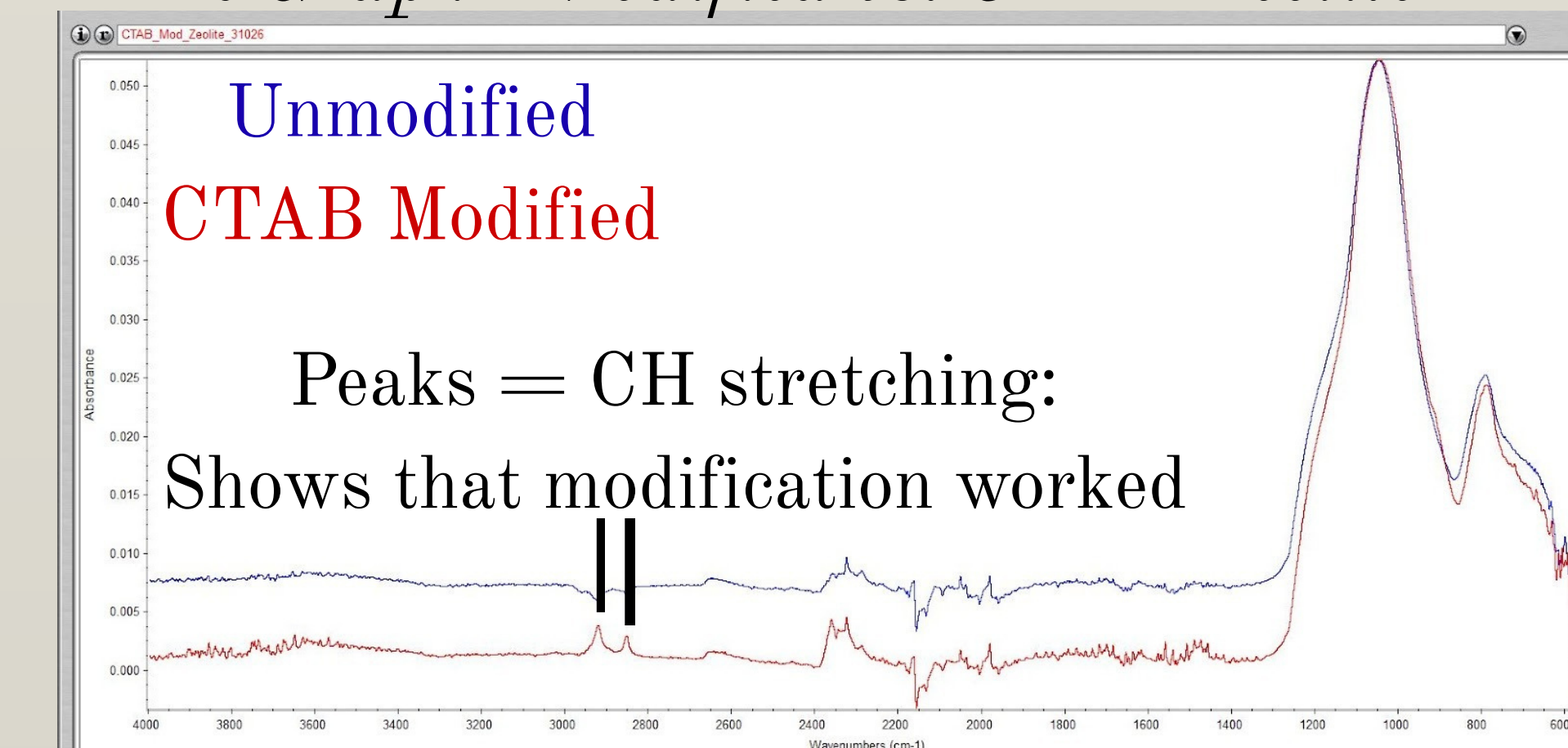
Results

Table - Each Treatment, Each Batch Acid Blue 25 Absorbance Data



Acid blue 25 solution absorbance:
Post-batch test solution absorbances should, ideally, be significantly lower than this

FTIR Graph - Modified vs. CTAB Zeolite



Picture - Modified Zeolite after Batch Test



Statistics + Conclusion

Using the Kruskal Wallis statistical test, it was found that:

- Modifying zeolite with surfactant successfully removed acid blue 25 for Batch 1 ($p < 0.05$).
- However, the modified zeolite does not maintain successful removal for Batch 2 ($p > 0.05$), hurting its reusability.
- Zeolite is mostly reusable ($p = 1$ between data for all zeolites' Batch 1 and 2).

These results support the conclusion that modifying zeolite with cationic surfactant initially improves removal of anionic pollutant acid blue 25 until saturation. As a result, the experiment does not fully support the widespread implementation of CTAB modification in zeolite filtration of acid blue 25. However, the limited amount of zeolite used in this experiment due to chemical availability/cost could have influenced results, especially zeolite reusability. Future research should focus on using higher quantities of zeolite and incorporating regeneration cycles.

References

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